Identifying historical geometries from complete rosters

Human Compression project: bounded source correspondence

1 A finite identification principle

Fix a finite group G, a rank, and a class of flag-transitive geometries with specified local hypotheses. A marked input is a family of actual subgroups (G_i). Two inputs are equivalent when one automorphism of G maps the whole family to the other after a permutation of types. The accepted shadow packets retain exactly this convention.

Proposition 1. *Suppose a complete roster for this class is given, together with a verified quotient into whole-family equivalence classes. Let Γ be a historical geometry known to belong to the class. Suppose its source gives the multiset s of type sizes and the order b of its chamber stabilizer. If exactly one roster class has these two invariants, then that class represents Γ with its stated G-action, up to an automorphism of G and a permutation of types.*

Proof. Choose a chamber in Γ and let G_i be its object stabilizers. Flag transitivity identifies the objects of type i equivariantly with the right cosets of G_i; hence their number is $[G : G_i]$. The chamber stabilizer is $\bigcap_i G_i$. A change of chamber conjugates the whole family, an isomorphism of the acting group applies one automorphism to it, and a change of type names permutes it. Therefore

$$\left(\text{multiset}\{[G : G_i]\}, \left|\bigcap_i G_i\right|\right)$$

is invariant on each whole-family class. Completeness puts the historical family into some roster class. It has the printed invariants, and the verified uniqueness makes this the specified class. This proof uses completeness as a premise; equality of aggregate numbers alone would not supply it. Nor does equality of invariants merge any two roster entries: the accepted exact transporters supply the family quotient. □

Proposition 2. *Suppose the same complete roster has exactly one class at rank r. Then every historical rank-r geometry satisfying its hypotheses has that class, regardless of its numerical name. Every isomorphism-invariant property established for this class holds for each such historical geometry.*

Proof. Apply completeness to the historical marked family. Its class must be the only class in the roster. The type permutation and group automorphism induce an incidence isomorphism, which transports every flag, residue, shadow and intersection witness. Therefore each of IP_k, WIP_k, full IP and full WIP has the same truth value. This uses exhaustive coverage of the class, not merely agreement between two reported counts. □

2 The unique rank-five Mathieu geometry

The retained complete phase-six M_{12} census exhausts all 147 Borel anchors and the larger class in which local two-transitivity is not assumed. It has exactly one rank-five class, represented by witness 55. Its Borel has order 3, its maximal parabolics have orders $108, 720, 720, 7920, 7920$, and it satisfies local two-transitivity. The exact prior source-to-census comparison identifies the

sole record of `m12.rk5` with witness 55. These are accepted computation premises, including the complete subgroup catalogue and finite group algorithms, not new computations or a new classification here.

Any M_{12} rank-five geometry denoted 5.1 in Tables 2 and 4 of [3] belongs to this larger class. The preceding proposition therefore identifies it without assuming that its suffix equals an archive position. The accepted all-residue computation proves IP_3 and IP_4 and disproves WIP_5. Hence both named positive claims hold, and the same geometry fails full WIP. All of these assertions use the exact specified M_{12} action and the retained Aut/type convention.

3 The negative rank-six Higman–Sims assertion

The unnumbered paragraph on printed page 289 of [3] names the rank-six HS geometry 6.1 and states that it fails WIP_6. This statement is separate from the 232 positive occurrences in its Tables 2, 4, 6 and 8. The retained earlier HS classification has exactly one rank-six class. The accepted quotient of the supplied roster represents it by `hs.rk6`, active entry 1, class 80 (global input 124). The singleton proposition identifies the historical geometry 6.1 with this family, under precisely that retained global completeness premise. Its accepted all-residue record has IP_2 but fails WIP_k for each $k = 3, 4, 5, 6$; in particular it verifies the named negative assertion. We neither infer completeness from the presence of one archive entry nor treat the geometry's historical name as an input to the shadow test. This adds one negative named assertion and does not change the count of four settled positive occurrences or the 228 remaining positive correspondences.

4 The Higman–Sims geometry numbered 3.2

The local primary article of Leemans [1], printed page 234, names its HoSi.c^* construction as geometry 3.2 in supplement [12]. Its displayed type sizes are $176, 15400, 1100$, and its Borel is S_6, of order 720. Reference [12] on page 236 identifies the same 2003 HS supplement and author URL as reference [21] of Jacobs–Leemans [3]. Thus the numeric name has an explicit documentary bridge; it is not inferred from its position in the supplementary permutation file.

The accepted HS computation processes all 124 active inputs and quotients them into 80 actual whole-family classes. It retains the historical classification as its completeness premise. Among all those inputs, the displayed three sizes and Borel select precisely the two rank-three active entries 2 and 3. Their subgroup orders are $2880, 252000, 40320$, in that order, in the group of order 44352000. The accepted transporter identifies them as the same class, class 2 with representative active entry 2. No other class has this invariant. By the proposition, under exactly that retained completeness premise, this is the historical geometry 3.2. Its accepted shadow computation has the full intersection property. Consequently the two occurrences of HS 3.2 in Tables 2 and 8 of [3] have an identified positive representative.

The second construction in [1], pages 234–235, has type sizes $176, 15400, 176$ and Borel order 720. These select class 1 uniquely, represented by active entry 1, whose accepted result is weak intersection without full intersection. The article body does not print its supplement number. We identify this *named construction*, but do not promote the tempting number 3.1 to an established historical label.

5 Hall–Janko diagrams and the exact limit of their names

The five-page primary article [2], pages 79–80, prints three rank-four diagrams as numbers 1, 2 and 3. Their invariants select the three representatives of the accepted complete Hall–Janko census as follows. The symbols R_{28}, R_{29}, R_{30} refer to its exact witness file, not to historical publication numbers.

Article diagram	Multiset of type sizes	Borel order	Witness
1	$4032, 4032, 10080, 10080$	1	R_{29}
2	$4032, 10080, 10080, 50400$	1	R_{28}
3	$100, 100, 1575, 1575$	8	R_{30}

Every line follows by dividing 604800 by the actual four subgroup orders and comparing the complete three-class roster. Thus diagram 2 has full intersection, whereas diagrams 1 and 3 fail weak intersection. No subgroup census or large shadow run is repeated here.

Jacobs–Leemans instead names a supplemental geometry 4.2. The already-local primary body does not explicitly identify its article ordinal 2 with that supplemental identifier. The common rank-prefix interpretation is plausible, but this packet keeps it as a documentary obligation. In particular these three identifications do not discharge any extra occurrence beyond the four settled claims among the 232 historical labels recorded in the frozen coherent L096 assembly.

6 What remains unresolved

The current fixed-access register still archives the Janko J_1 classification chapter of Gottschalk–Leemans and the related Magma-classification article as unavailable. The accessible thin-J_1 and Livingstone papers do not give the needed historical number 4.1. The existence of one positive and one negative rank-four class therefore does not identify that name. No new retrieval was attempted.

The files called `catalogue_L097.pdf`, `catalogue_L098.pdf` and `catalogue_L101.pdf` have exactly the same hash and body as the Hall–Janko article [2]; they are not primary sources for J_3, M_{23}, M_{22}. We preserve this previously known access limitation. The archive README explicitly identifies `hs.rk4`'s final `geo[2]` as 4.1 of a cited reference, whereas the accepted calculation proves that exact input fails both IP_3 and IP_4. Its literal reference number is [20], but the final paper's HS supplement is [21]. Thus this is an internally inconsistent edition-level routing assertion when combined with the positive table claims. We record the conflict without deciding whether the example, reference number, external label or table was misprinted. No other label assignment follows from it.

This additive note identifies HS 3.2 and the unique M_{12} rank-five class named 5.1, proving four positive occurrences and leaving 228 positive-label occurrences without an established historical correspondence. It does not complete L096, recover any additional original, or claim a new classification, theorem priority or length factor. All accepted common proofs and complete program/data dependencies remain paid; this finite invariant comparison does not replace them.

References

[1] D. Leemans, The residually weakly primitive geometries of HS, Australas. J. Combin. 33 (2005), 231–236. Exact local primary: `older_L091.pdf`; identifiers read directly on printed pp. 234–235.

[2] D. Leemans, The residually weakly primitive geometries of J_2, Note Mat. 21 (2002), 77–81. Exact local primary: `catalogue_L104.pdf`; diagrams on printed pp. 79–80.

[3] P. Jacobs and D. Leemans, An algorithmic analysis of the intersection property, LMS J. Comput. Math. 7 (2004), 284–299. The source's Tables 2 and 8, and reference [21], are used here.